Symbolic Music Generation

A causal Transformer trained from scratch on POP909

- Student ID: r14942154
- Code, own model weights and MIDI/WAV results:
- https://github.com/alexsu30812-gif/hw2
- Basic a: 5 settings x 20 complete 32-bar pieces; submit the selected 20 MIDI/WAV pairs.
- Basic b: all 9 prompt continuations. Bonus: all 10 melody-conditioned accompaniments.
- Extra experiments: context length, token prediction, quantization, sampling, checkpoint learning and melody conditioning.

Hosting note: GitHub substitution for the slides' cloud-drive wording is unverified; no TA acceptance is claimed.

Main results and how to read them

Task	Measured result	Submitted listening files
Basic a	t12: H4 2.7846; GS 0.970166	20 selected MIDI + 20 WAV
Basic b	Best by local boundary rule: 001:t12, 002:t12, 003:t10	9 MIDI + 9 WAV; original 8 bars retained
Bonus	10 prompts; required CA and four OA measures reported	10 MIDI + 10 WAV; original melody retained

- The report compares actual saved outputs. All final checkpoints have observed the complete 896-song training set; no pretrained weights or extra training data are used.

- A 45-song split supports pre-refit diagnostics. Those songs enter the final all-song fit, so the real-data reference is descriptive, not an independent test of the final model.

- H4/GS and CA/OA are proxies. No human preference ratings were collected, and none of these values is a course-grade formula.

Data preparation: trust the beat annotations

- Use only the supplied 896 training songs. Merge MELODY, BRIDGE and PIANO for basic tasks; retain separate MELODY/PIANO roles for accompaniment. All 13 official prompt songs are excluded from fitting.

- Song-level split before window sampling: 851 optimization songs, 45 validation songs; seed 20261022. The 13 official prompt folders are excluded.

- The TA announcement warns that many source MIDI time signatures are wrong. Bar boundaries and beat times come from beat_midi.txt; source MIDI time-signature metadata is not authoritative.

- Canonical export: one annotation beat becomes one quarter note; a bar with n supplied beats is written as n/4. This preserves the supplied clock and bar count, including mixed meters. It does not recover an original denominator such as 6/8.

- Short pickup bars are retained. Notes before the first annotated beat are clamped; representation error is measured rather than called lossless. Original prompt notes are restored after generation.

Token representation and causal model

REMI-style event sequence

BOS, [Bar, Meter, Tempo, (Pos, Pitch, Vel, Dur)*], ..., EOS

683 tokens; 8 subdivisions per supplied beat; 32 velocity bins; note duration 1..256 grid steps (up to 32 beats).

For bonus, each bar contains Track_Melody + its notes, then Track_Piano + its notes. Future-bar melody is not visible while that bar's piano is sampled.

Decoder-only Transformer

6 layers; width 384; 6 attention heads; 12,482,688 parameters.

Learned absolute positions (4096 capacity), pre-layer normalization, GELU, dropout 0.1 and tied embedding/output weights. Selected effective training/inference context: 2048 tokens.

Train by next-token cross-entropy, ignoring padding. Causal attention sees only preceding tokens. All music-model weights originate from random initialization.

Reliable decoding uses legal event-type masks, ascending pitches at one onset, piano-range pitches 21..108, a 64-note/bar cap and exact end boundaries. Overflow retains recent complete bars and rebuilds the KV cache; this limits long-range memory.

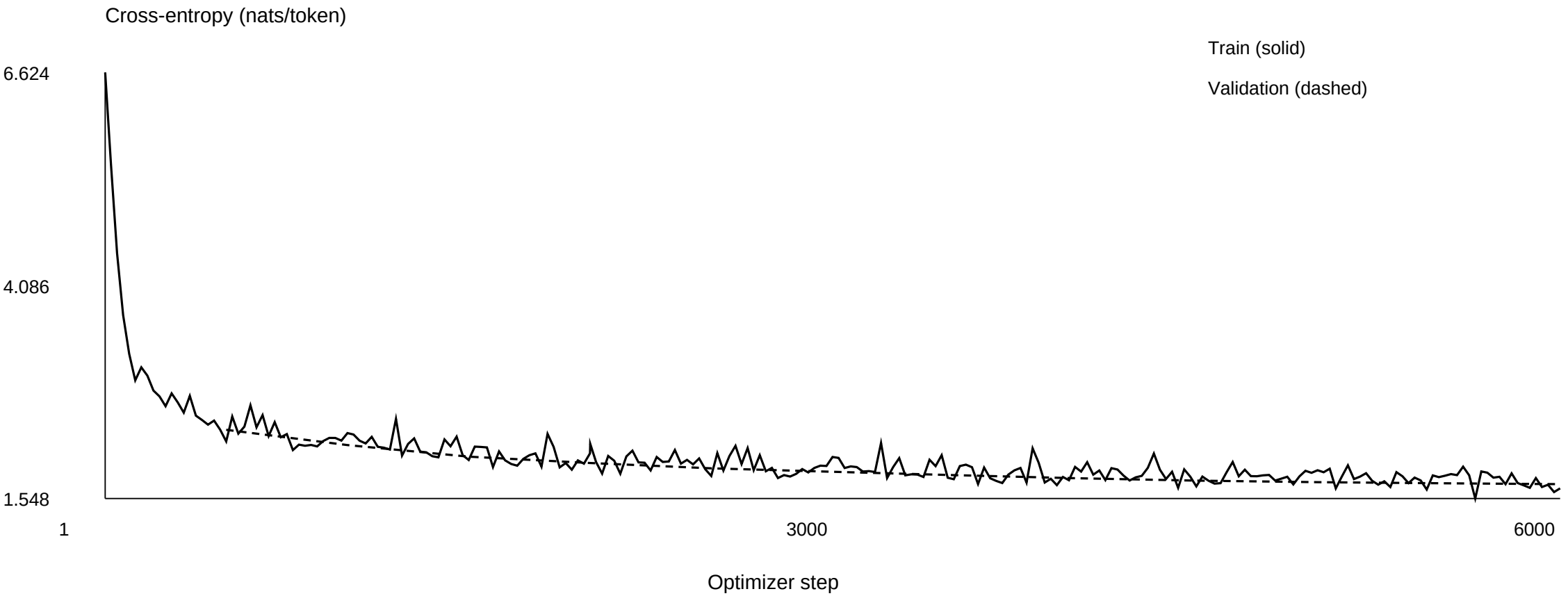
Optimization and checkpoint lineage

Apple M4 Pro / PyTorch MPS training. Own checkpoint continuation resets the optimizer at each stage.

Stage	Steps	New targets	Context	Songs seen	Val CE
basic_diagnostic	2000	13.751M	2048	851	1.9712
basic_refined	4000	27.544M	2048	851	1.7192
context_2048	579	4.007M	2048	851	1.7064
context_4096	696	4.003M	4096	851	1.7408
basic_all	4081	28.006M	2048	896	not held out
roles_diagnostic	1194	8.003M	2048	851	1.5046
roles_all	1794	12.005M	2048	896	not held out

AdamW: betas (0.9, 0.95), weight decay 0.1, gradient norm clip 1.0; warmup + cosine decay. Learning rates by row: 3e-4, 1.8e-4, 8e-5, 8e-5, 5e-5, 1.5e-4, 5e-5. Batch 4, except context_4096 batch 2. Legal pitch transposition -5..+5 semitones; no validation augmentation.

Training and validation loss



The fixed 851/45 song split uses deterministic 2048-token validation windows. The optimizer restarts after step 2000. CE falls from 2.36666 at step 500 to 1.71925 after refinement. Final all-song fitting has no held-out validation score.

Extra experiment: does more context help?

Two branches start from the same refined checkpoint SHA256. Both use AdamW, learning rate 0.00008 and a 4,000,000 non-padding target budget. Compare last checkpoints on the same 45-song, 2048-token validation windows; allow at most one training-step budget overshoot.

Context	Target tokens	Batch	Validation CE	Seconds
2048	4006540	4	1.70637	309.10591
4096	4003086	2	1.74084	591.45663

- The 2048-token branch has lower validation CE (1.70637 versus 1.74084) and is selected before the all-song final fit. The measured token-budget difference is 3,454 targets (0.086% of the nominal budget).

- One run per context, with different batch sizes and step counts. The shared parent was trained mainly at 2048 tokens with learned absolute positions; the 4096 branch must adapt later positions. Validation also uses 2048 tokens. This compares this warm start and budget, not the general value of long context.

Extra experiment: grammar versus note content

Teacher-forced full-vocabulary prediction with no grammar mask: 77,926 non-padding targets from the same 48 deterministic windows covering all 45 pre-refit validation songs. Window hashes match across the step-500 and step-2000 checkpoints.

Target family	CE @500	CE @2000	Acc % @500	Acc % @2000
Tempo	3.848	0.543	4.176	87.692
Pos	1.361	0.841	52.124	67.175
Pitch	3.272	2.654	8.713	19.304
Vel	2.323	2.201	20.768	24.451
Dur	2.583	2.395	21.016	23.982

Overall CE falls from 2.36666 to 1.97120 nats/token and top-1 accuracy rises from 26.54% to 35.51%. Token-family accuracy is already 98.30% at step 500; pitch-content accuracy rises from 8.71% to 19.30% and remains much harder than recognizing token grammar. Teacher forcing supplies the correct history and does not measure free-running musical quality. These checkpoints were trained on the 851-song optimization split; the final all-song models are not claimed to have held-out validation accuracy.

Basic a: sampling settings and required metrics

Each setting uses the same final checkpoint, CPU float32 and piece seed 20261022+i (i=0..19). Means +/- piece standard deviations.

Set	Sampling	H4	GS (official)
Real reference (45)	same adapter	2.6398 +/- 0.1225	0.970831 +/- 0.013551
t08	T=0.8, p=0.95	2.6072 +/- 0.1202	0.997488 +/- 0.002611
t10	T=1, p=0.95	2.7255 +/- 0.1197	0.992609 +/- 0.004625
t12 [selected]	T=1.2, p=0.95	2.7846 +/- 0.1132	0.970166 +/- 0.024619
k16	T=1, k=16	2.7256 +/- 0.0571	0.981920 +/- 0.007164
k32	T=1, k=32	2.7061 +/- 0.1117	0.982149 +/- 0.009521

H4 is pitch-class entropy over 4-bar windows; GS compares bar onset patterns. Unconditional generation fixes 4/4 and 120 BPM. All 100 pieces pass 32-bar/ID/hash/note checks with zero skips. Real/generated music uses the same Pos8-to-Pos16 mapping; 2000 whole-piece bootstrap draws provide saved confidence intervals.

Why select t12? Inspect density and rhythm too

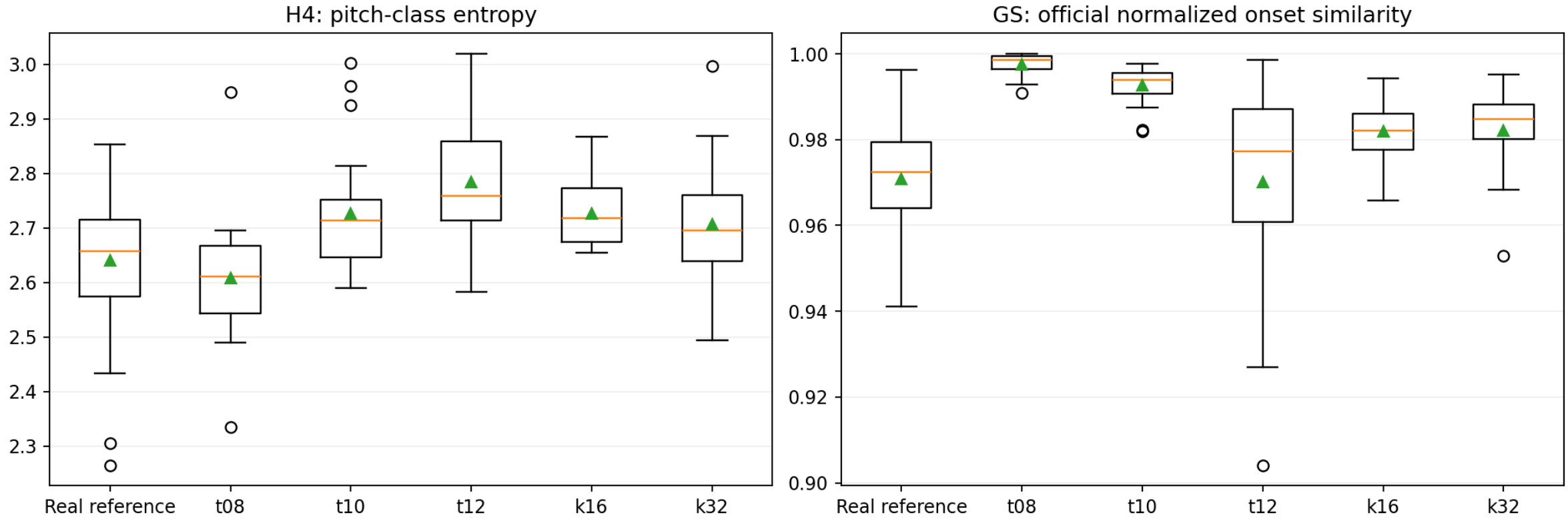
Set	Notes/bar	GS active	Velocity SD	Mean duration/beat
t08	38.952	0.9900	3.833	0.448
t10	33.080	0.9704	6.757	0.508
t12	21.045	0.8807	13.236	0.816
k16	26.050	0.9277	12.039	0.646
k32	23.859	0.9286	15.497	0.744

- Predeclared choice: minimize the mean of absolute H4 and GS mean gaps divided by real-reference SDs. t12 scores 0.6155, versus k32 0.6882 and k16 0.7593. This rule selects on the same reported samples; it is not an independent test.

- t12 has 21.045 notes/bar versus real 22.953. T=0.8 is much denser (38.952), with nearly identical onset patterns (GS active 0.98995) and little velocity variation (SD 3.833). Exact note-level bars are still unique; the collapse is rhythmic/expressive, not literal repeated bars.

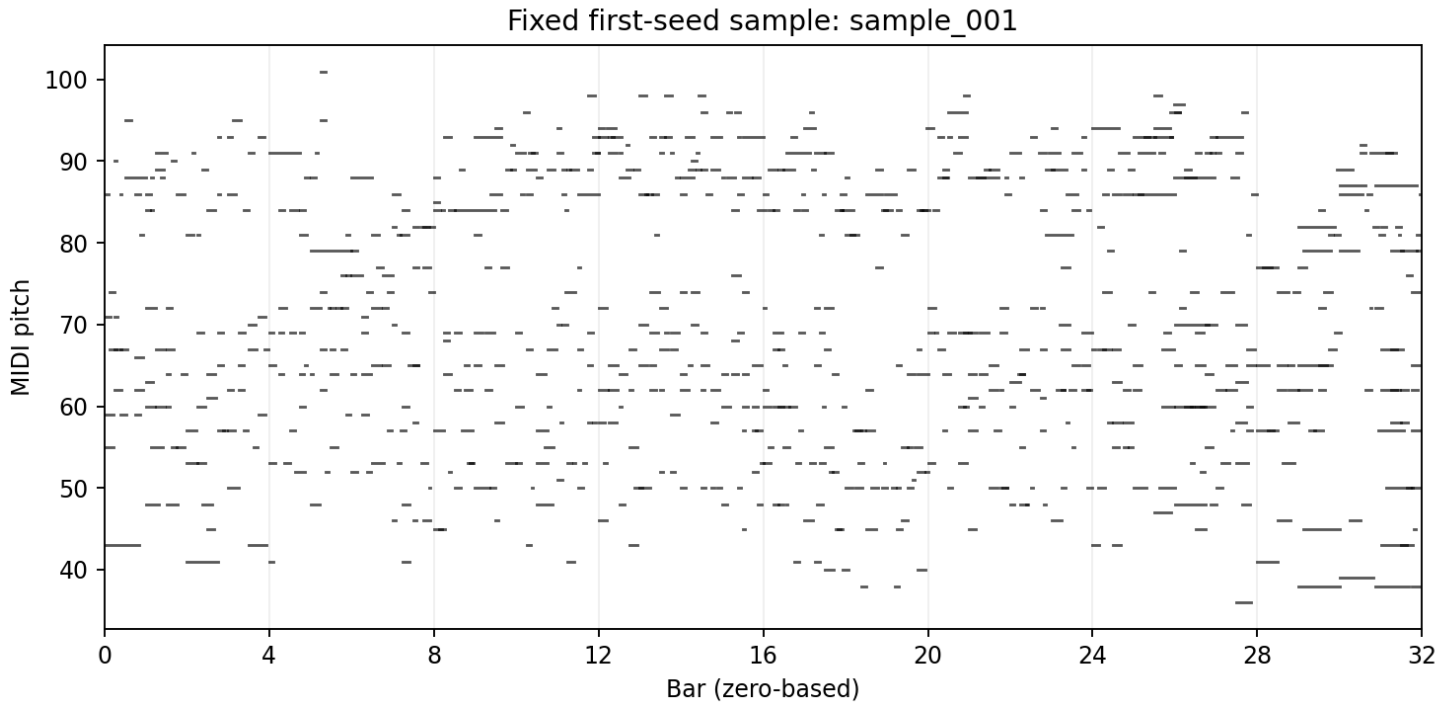
- t12 minus t10: GS -0.02244, paired 95% bootstrap CI [-0.03428, -0.01233]. H4 +0.05914, CI [-0.01600, 0.13447], so the H4 difference is less certain. No human listening preference is inferred.

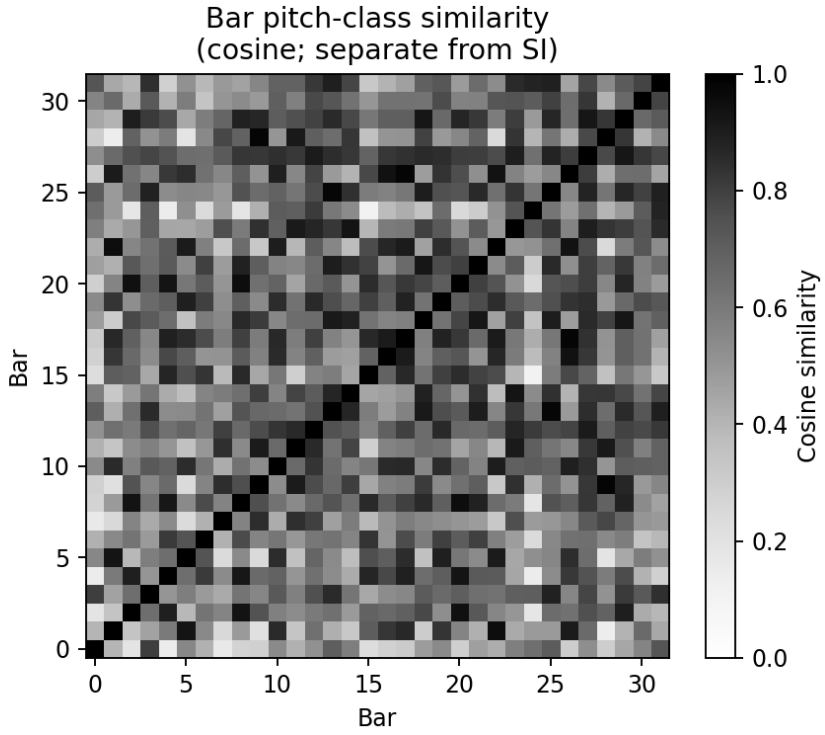
Sampling changes distributions, not just one mean



All 20 pieces per setting are shown. The real reference uses the same event grid. These are metric distributions, not listener scores.

One generated piece: notes and bar similarity





t12 / sample_001: the first saved seed, not a cherry-picked example. Cosine similarity of bar pitch-class counts is a diagnostic; this is not the optional MusDr SI score.

Basic b: all 9 continuations and chosen versions

Supply 8 original bars, sample 24 new bars, then restore every original prefix note. All nine retain the 8-bar note identity and contain new notes.

Prompt	Set	Sampling	Boundary score	Density ratio	Pitch jump	Choice
001	t08	T=0.8; p=.95	1.0342	1.926	4.546	
001	t10	T=1; p=.95	1.0499	1.815	3.733	
001	t12	T=1.2; p=.95	0.4552	1.333	-0.296	BEST
002	t08	T=0.8; p=.95	0.8008	1.239	1.037	
002	t10	T=1; p=.95	0.7137	1.522	2.026	
002	t12	T=1.2; p=.95	0.5566	0.913	2.588	BEST
003	t08	T=0.8; p=.95	0.4288	1.113	-2.144	
003	t10	T=1; p=.95	0.3285	1.113	0.941	BEST
003	t12	T=1.2; p=.95	1.0464	1.226	7.108	

Selection compares the two bars before/after the boundary: |log density ratio| + |mean pitch jump|/12 + (1 - pitch-class Jaccard). Lower wins within each prompt. All use the same basic checkpoint and CPU; only T=0.8/1.0/1.2 changes. Best labels are local continuity preferences, not listener judgments.

Continuation: what the chosen labels establish

- 001 / t12: boundary density ratio 1.333, mean pitch shift -0.296 semitones, pitch-class Jaccard 0.857. Its score 0.4552 is lower than t08 1.0342 and t10 1.0499; the other settings nearly double the local density.

- 002 / t12: density ratio 0.913, pitch shift +2.588, Jaccard 0.750; score 0.5566 versus 0.8008 and 0.7137. It preserves local density better, although it does not minimize every descriptor separately.

- 003 / t10: density ratio 1.113, pitch shift +0.941, Jaccard 0.857; score 0.3285. t12 instead jumps +7.108 semitones at the seam, illustrating that the best unconditional temperature need not be best for every prompt.

- Boundary continuity does not evaluate the remaining 24-bar form. The model sees only its recent complete bars after cache rollover. Full-32-bar and continuation-only-24-bar metrics are saved separately; the supplied prefix is not counted as newly generated music.

Bonus: melody-conditioned piano generation

- Start from our own refined basic checkpoint, then learn MELODY/PIANO bar interleaving on 851 songs; refit on all 896. This is an own-weight transfer stage, not use of a pretrained external model.

- For each of 16 bars, insert that bar's melody tokens and sample its piano events. The model sees past bars and the current melody bar, not the full future melody. BRIDGE is excluded from role training.

- Generation reads only melody.mid and beat_midi.txt. Ground-truth piano and chord annotations are evaluation-only, so generated piano is not given the reference accompaniment or the target chords.

- Formal bonus configuration: {"bars": 16, "conditioning": "normal", "temperature": 1.0, "top_k": 0, "top_p": 0.95}. The original melody pitch/velocity is retained, with sub-millisecond export rounding checked independently. Exactly MELODY + PIANO tracks and all 16 bars are validated.

- The source melody is restored after token generation, including mixed-meter prompts and the long ending of prompt 007. Generated piano uses bounded durations and no active same-pitch overlap.

Bonus: all 10 prompt results

Each row evaluates all 16 bars. OA_P is the supplied pitch-class overlap measure; higher overlap alone does not establish harmonic correctness.

Prompt	CA	OA_P	OA_V	OA_D	OA_IOI
001	0.1250	0.8628	0.1864	0.7794	0.6378
002	0.0625	0.8356	0.5613	0.6928	0.6695
003	0.1875	0.8577	0.6127	0.6806	0.5059
004	0.0000	0.8572	0.5564	0.5609	0.2987
005	0.1250	0.8577	0.7719	0.7475	0.4433
006	0.1250	0.8464	0.6240	0.7007	0.4136
007	0.0625	0.8775	0.7011	0.8512	0.7225
008	0.1250	0.8549	0.7008	0.7925	0.5936
009	0.2500	0.8423	0.7084	0.7376	0.8253
010	0.0625	0.8599	0.7080	0.8150	0.5772

Means over all ten prompts: CA 0.1125; OA_P 0.8552; OA_V 0.6131; OA_D 0.7358; OA_IOI 0.5687. Zero skipped files. CA-ceiling 0.7625 is the same chord estimator on ground truth, not a strict upper bound.

Bonus: select the inference setting

Hold the roles checkpoint, ten melody prompts, seed schedule and top-p .95 fixed; compare normal-conditioning temperatures .8, 1.0 and 1.2. Select the largest equal-weight average of the five official CA/OA metrics.

T	CA	OA_P	OA_V	OA_D	OA_IOI	Mean
0.8000	0.1000	0.8557	0.4559	0.5948	0.5151	0.5043
1.0000	0.1125	0.8552	0.6131	0.7358	0.5687	0.5771
1.2000	0.1437	0.8602	0.5811	0.6724	0.6055	0.5726

The predeclared equal-weight composite selects T=1.0 (0.57708), narrowly above T=1.2 (0.57258). T=1.2 has higher CA and IOI overlap, but lower velocity/duration overlap. We retain T=1.0 for the formal files and conditioning ablation; no setting dominates every descriptor.

All three settings use the same ten prompts and evaluation ground truth: this is in-sample tuning, with no independent test. The custom composite is not the course grading formula and does not establish listener preference. All per-prompt scores and paired bootstrap differences are retained.

Extra experiment: remove melody conditioning

Same roles checkpoint, ten prompts, seed schedule and T=1.0 / top-p=0.95. Remove melody-note tokens while keeping bar, meter, tempo and track headers; restore the original melody in both exported versions. Evaluate official CA/OA and bootstrap 5000 paired whole-prompt resamples.

Metric	Normal	Empty melody	Normal-empty	95% low	95% high
CA	0.1125	0.0375	0.0750	0.0250	0.1313
OA_P	0.8552	0.8504	0.0048	-0.0034	0.0128
OA_V	0.6131	0.3779	0.2352	0.1359	0.3382
OA_D	0.7358	0.6002	0.1356	0.0343	0.2615
OA_IOI	0.5687	0.4933	0.0754	-0.0976	0.2574

Normal melody improves CA by 0.0750 (95% CI [0.0250, 0.1313]), velocity overlap by 0.2352 and duration overlap by 0.1356. Pitch-class and IOI intervals include zero. This provides evidence of a conditioning effect on measured descriptors, with weak evidence for some metrics.

Caution: deleting all melody is out of distribution and changes context length/sampled histories. Both exports restore the full melody; the empty condition is not submitted. This does not isolate an effect on perceived musical quality.

Discussion, limits and reproduction

- Useful changes: more training improves note-content prediction; the controlled 2048-token branch predicts better at this budget; warmer sampling reduces the dense, nearly fixed onset behavior of low temperature.

- What remains weak: accompaniment CA is low even when pitch-class OA is high. The model has no explicit chord input or harmonic objective. Distribution overlap can hide wrong chord order, register or voice leading; a high OA_P should not be read as correct harmony.

- Long-range form is unverified. Absolute-position resets, finite recent-bar context and grammar constraints can limit phrasing. No human listening study, architecture sweep or official SI experiment is claimed.

- Reproduce inference with pip install -r requirements.txt, then src/infer.py; README gives exact checkpoint, seed, settings, output-path and WAV-rendering commands. All 39 submitted MIDI/WAV pairs are synthesized from matching MIDI and checked for duration, finite audio and non-silence.

- The public package includes two own checkpoints, vocabulary, code, licenses, measured tables/manifests, figures and hashes. It excludes training MIDI, preprocessed training arrays, external weights and caches. See the appendix for audit experiments and references.

Appendix A: quantization and export error

All 896 original training songs; identical annotation beat maps, rounding, velocity bins, merged tracks, 32-beat maximum duration. Position vocabulary supports 16 beats per bar at every resolution. No Transformer is retrained for this ablation.

Steps/beat	Vocabulary	Onset MAE ms	Offset MAE ms	Grid collisions
4	491	5.692	50.810	45779
8	683	3.192	23.956	44076
16	1067	1.747	12.082	42856

- Across 1,508,536 notes, 16 subdivisions lowers onset/offset MAE but increases vocabulary from 683 to 1067 tokens. We implement 8; this is a reconstruction audit, not a comparison of models retrained at each grid.

- Six codec round-trip cases preserve all represented notes and have maximum MIDI export timing error below 0.317 ms. This small export error is separate from source-to-grid quantization error.

- The duration cap clips two long source notes (worst offset error 8.732 s); 196 early notes are clamped before the first annotation (worst onset error 0.709 s). The representation is therefore not lossless. Full per-case audits and reproduction code are provided.

Appendix B: checkpoint progress and GS sensitivity

Checkpoint	H4	GS	GS active	Notes/bar	Empty fraction
checkpoint_0500	2.9303	0.9707	0.8829	11.5594	0.0328
checkpoint_2000	3.0407	0.9898	0.9592	28.1641	0.0000

- At step 500, GS 0.97074 already resembles the real reference 0.97083, but density is only 11.559 notes/bar versus 22.953 and 3.28% of bars are empty. At step 2000, density rises to 28.164 while H4/GS move away from the real means. Lower prediction loss alone does not optimize these descriptors.

- The required official GS uses 128 binary positions. Mapping our 8 subdivisions to the official 16-subdivision IDs leaves odd slots zero. GS active separately normalizes onset XOR by max(bar beats) x 8; it does not replace official GS.

- For the real reference GS/GS active are 0.97083/0.88679; selected t12 gives 0.97017/0.88066. Shared grids make the comparison fair, but onset-only similarity ignores pitches, durations, velocities and the number of simultaneous notes.

- Checkpoint generation uses completed 20-piece sets. These pre-refit checkpoints are diagnostics; context/learning stage effects are not an isolated controlled quality comparison.

Appendix C: reference sensitivity and local matches

Take one deterministic center-nearest eligible 32-bar crop per song using the same official metric adapter. Consider all 896 training songs; list individually the five with no eligible crop. Keep the original 45-song reference and configuration-selection rule fixed.

Reference	Included	Excluded	H4	GS	Notes/bar
Primary45	45	0	2.6398	0.9708	22.9528
All eligible	891	5	2.6407	0.9706	22.4902

- The all-eligible reference includes 891/896 songs; five songs have no eligible 32-bar crop. Its H4/GS/density means fall inside the 45-song primary-reference bootstrap intervals. The original reference and t12 choice are retained; the two reference sets overlap and are not independent tests.

- No duplicate full pieces and no exact non-empty 1/2/4/8-bar training-window matches are found in any final basic-a setting. Windows match position, pitch and duration while ignoring tempo/velocity/track names. Shared fully silent windows are reported separately.

- This narrow exact-match check cannot rule out transposition, approximation, partial copying or memorization. Zero exact matches is not proof of originality. Selection and uncertainty estimates use reported output sets, not a new hidden test.

References, source evidence and AI disclosure

- Bo-Rui Chen. Course Homework 2: Symbolic Music Generation, 2026. Task requirements pp.11-18,22-28; submission pp.29-32; grading p.33; AI permission p.34; rules p.35.

- Z. Wang, K. Chen, J. Jiang, Y. Zhang, M. Xu, S. Dai, X. Gu and G. Xia. POP909: A Pop-Song Dataset for Music Arrangement Generation. ISMIR, 2020. https://arxiv.org/abs/2008.07142

- Y.-S. Huang and Y.-H. Yang. Pop Music Transformer: Beat-based Modeling and Generation of Expressive Pop Piano Compositions. ACM Multimedia, 2020. https://arxiv.org/abs/2002.00212

- Y. Ren, J. He, X. Tan, T. Qin, Z. Zhao and T.-Y. Liu. PopMAG: Pop Music Accompaniment Generation. ACM Multimedia, 2020.

- Course HW2_scripts/tokenization.py, eval_basic.py and eval_bonus.py; upstream MusDr side_utils.py: https://github.com/slSeanWU/MusDr/blob/master/musdr/side_utils.py

AI assistance: Codex assisted implementation, execution, result aggregation, analysis and English drafting. The HW2 slides permit internet and AI use (p.34). All reported numbers come from saved actual outputs. Full SHA256 inventories, checkpoint ancestry and per-piece tables are in results/report_data.json, FILE_SHA256.json and SUBMISSION_CHECKS.json.